HUMAN-ROBOT INTERACTION · FINAL PRESENTATION

BrewBot

A Physical-State-Aware Drink-Providing Robot

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The BrewBot Concept

Kitchen drink-service is low-stakes but physically meaningful	HRI task with real sensing, dialogue, and movement
Reduce participant burden	Robot should notice state changes and offer help instead of waiting for explicit commands
Concrete evaluation scenario	Every step can be tested: sensing, decision, conversation, arm motion, and drink preparation

Sense
Smartwatch

→

Infer
State Estimation

→

Interact
Gestures

⇄

Act
Serve

Original Project Idea

Planned input	HR, HRV, EDA, skin temperature, speech, gestures, time/context
Planned output	Drink suggestion, confirmation dialogue, robot arm movement, coffee/water preparation
Design goal	A modular system where inference decides and reusable actuator nodes execute

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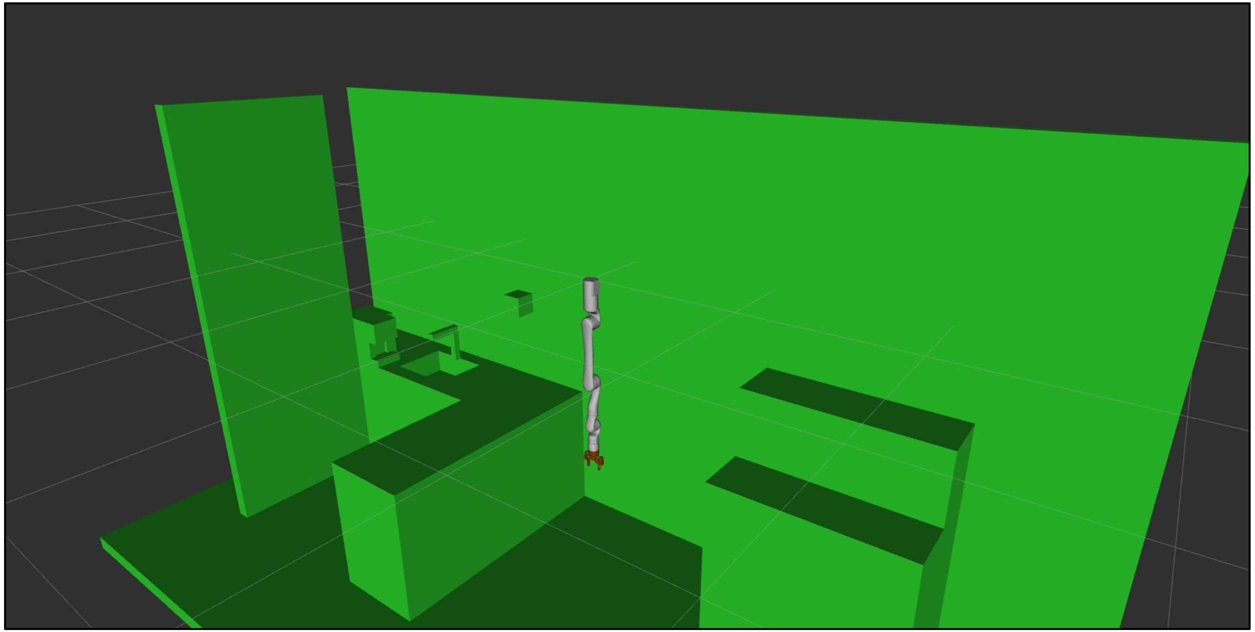
Pre-Hackathon

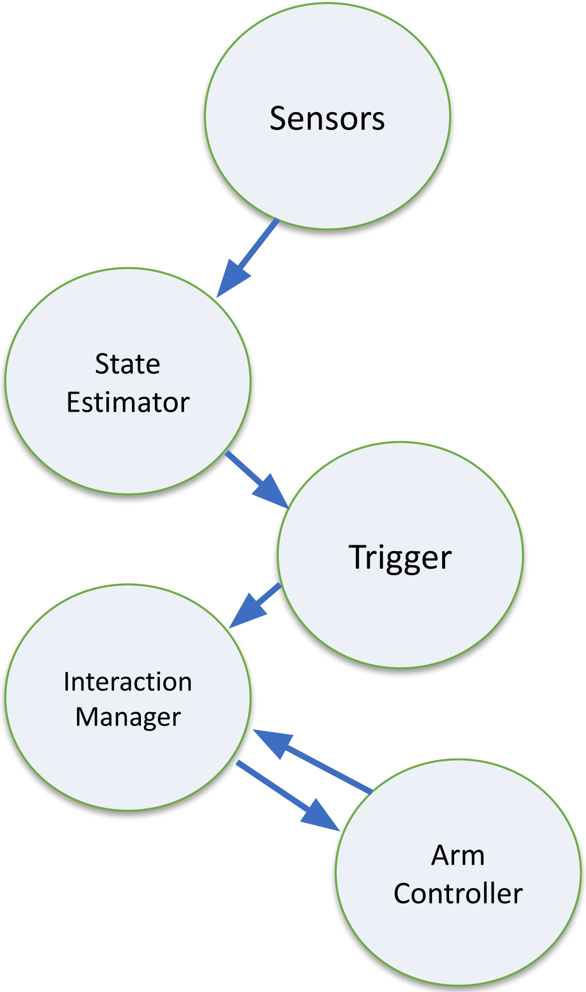
- Learning arm movement and deciding to use MoveIT
- 3D-modelling kitchen for collision detection
- Designing and modelling cup for later 3D-print



Architecture

Sense → Decide → Interact → Act

- Core Idea: Each node owns one responsibility. State inference, dialogue, movement, and devices stay separate
- Replaceability: benefits from ROS2's modular design. Easy to swap components.
- Trigger policy: periodic and reactive to state spikes



How The System Grew

0. Groundwork	Arm movement via MoveIt, 3D modelling kitchen
1. Interaction v1	Interaction manager, TTS, speech input
2. Sensors	Wearable (E4 + Smartwatch) for physiological data
3. Drink Workflows	Arm drink workflows, coffee machine via Home Assistant
4. Interaction v2	LLM-backed confirmation and reply classification, visual light feedback
5. Perception	Gesture and camera work toward dynamic interaction and cup localization

Brewbot At Runtime

Interaction

- Natural Speech processing
- Cameras and Gesture Detection
- Light control as additional indicator
- Humanoid motion

→

Decision

- Rule-based state estimator
- Trigger node
- Decides when/how to serve

→

Robot Actions

- MoveIt arm control: End pose hard coded. Path to it is not.
- Rail, lift, arm, gripper,
- Home assistant integration: Robot as ecosystem

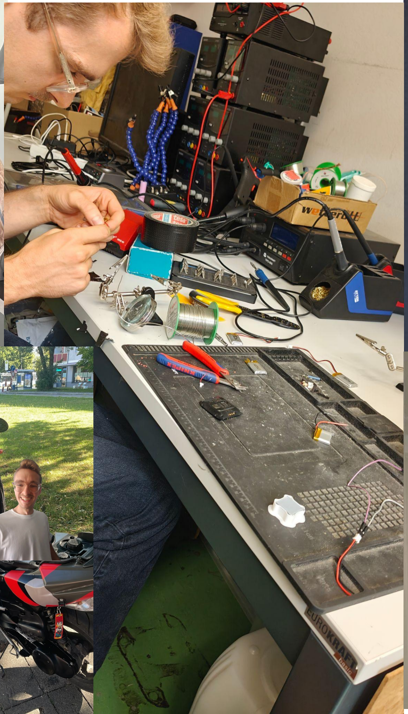
IMPLEMENTATION HISTORY

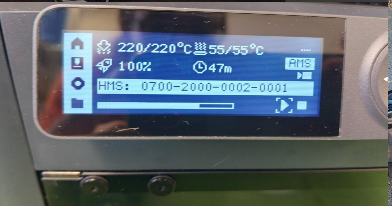
Gestures And Cameras

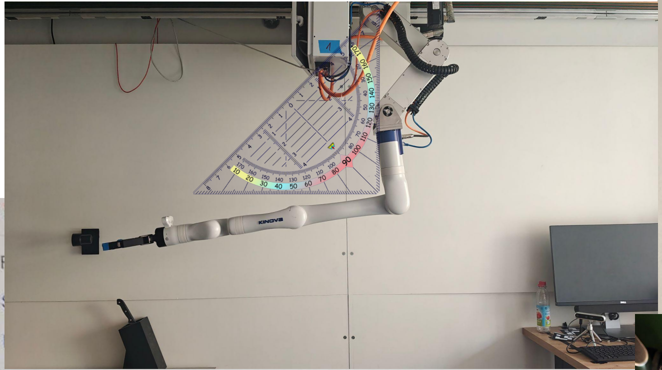
- Gesture recognition: camera/gesture topics and arm-camera gesture integration
- Interaction channel: Gestures complement speech and physiological sensing for non-verbal interaction
- Two-sided gestures: Robot can mime to propose drinks, user can answer with gestures

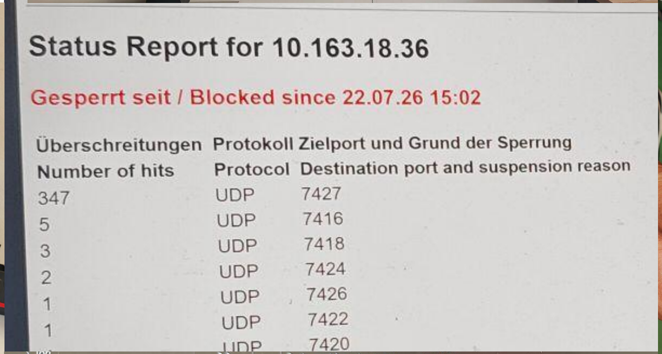
Conclusion

Difficulties and insights









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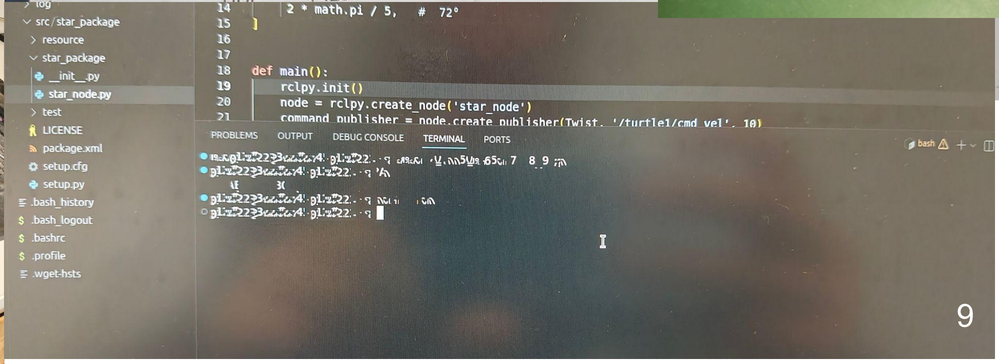

What Comes Next?

Dynamic cup localization

- AprilTags/camera perception so the arm moves to cup location
- Direct handover to user

More interaction modes

- gestures
- speech commands
- proactive state-triggered interactions

More drinks and robustness

- more validated drink programs
- better sensor-calibration/ per-user preference tables

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BrewBot Became A Modular HRI Pipeline

From physiological cues and interaction
to
physical drink-serving behavior

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7:30
Jogging